



RS003 Week 6 — English Worksheet

Topic: A Fleet of Ships — Many Parameters · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week is about thinking and talking about code, not typing it in the app. One function makes a whole **fleet of different ships**: you give `draw_ship` more parameters — position, colour, and size — so a single function can draw a small green scout, a big red boss ship, and a blue one, all at once. You will predict, debug, and explain code, then explain the code you wrote in today's lesson.

Words to know: **parameter, argument, default value, keyword argument, scale (size)**

1 · Predict 🌟

Read each piece of code. Do not run anything — just write what you think will happen.

```
def draw_ship(x, body_color, size=1):
    ...
```

This function takes three parameters. Which one has a default value, and what does that mean when you call the function?


```
draw_ship(200, GREEN, size=0.8)
draw_ship(400, RED, size=1.2)
draw_ship(600, BLUE, size=0.8)
```

Three ships are drawn. Which one is the biggest? Which one is in the middle?


```
w = int(40 * size)
```

If `size` is 1.2, what does `int(40 * size)` work out to? Why is `int(...)` needed?


```
draw_ship(400, RED)
```

Here `size` is left out. What size will this ship be, and how do you know?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken.

Bug A — This should draw a ship in any colour, but every ship comes out the same colour no matter what you pass in.

```
def draw_ship(x, body_color, size=1):  
    pygame.draw.rect(screen, GREEN, (x - 20, HEIGHT - 120, 40, 50))
```

Hint: the function ignores its own `body_color` parameter.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This should make a ship 1.2 times bigger, but Python crashes with an error about needing a whole number.

```
w = 40 * size  
pygame.draw.rect(screen, body_color, (x, HEIGHT - 120, w, 50))
```

Hint: `40 * 1.2` is a decimal. Pygame sizes must be whole numbers.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — You want most ships to stay normal size without typing `size=1` every time, but right now leaving size out causes an error.

```
def draw_ship(x, body_color, size):  
    ...  
  
draw_ship(400, RED)
```

Hint: give `size` a default value so it can be left out.

Write the fixed code (just the `def` line):

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working example carefully. You do not need to run it — just answer the questions about how it works.

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

SPACE = (10, 12, 40)
WHITE = (255, 255, 255)
GREEN = (60, 200, 120)
RED = (230, 60, 60)
BLUE = (80, 140, 255)

def draw_ship(x, body_color, size=1):
    w = int(40 * size)
    h = int(50 * size)
    nose = int(20 * size)
    y = HEIGHT - 120
    pygame.draw.rect(screen, body_color, (x - w // 2, y, w, h))
    pygame.draw.polygon(screen, WHITE, [(x, y - nose), (x - w // 2, y), (x + w // 2, y)])

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    screen.fill(SPACE)
    draw_ship(200, GREEN, size=0.8)
    draw_ship(400, RED, size=1.2)
    draw_ship(600, BLUE, size=0.8)
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

What are the three parameters of `draw_ship` , and which one has a default value?

Look at `w = int(40 * size)` . Why does a bigger `size` make a wider ship?

In `draw_ship(400, RED, size=1.2)` , name each argument and say which parameter it goes to.

Why can the same `draw_ship` function draw three different ships? What changes each time?

The `pygame.draw.polygon` line draws the white nose. How does it use `size` to stay the right shape?

4 · Explain Your Lesson Code 🎥

In today's live lesson you wrote your own ship-fleet code. Record a short video on your phone explaining that code in your own words. You may show your fleet running on screen. Try to use these words: **parameter**, **argument**, **default value**, **keyword argument**, **scale**.

1. Show your row of ships.
2. Read one of your function calls out loud and name each argument.
3. Explain what the default value of `size` does in your code.
4. Point at your boss ship and say what makes it bigger.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.